Ladder Lessons Website UAT Report

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Overview

This report details the User Acceptance Testing (UAT) performed on the Ladder Lessons website deployed on GitHub Pages (URL: `https://sharichung.github.io/ladder-lessons/`). The testing aimed to verify the functionality, usability, and overall quality of the website, including page loading, navigation, interactive elements, and responsiveness.

Test Cases and Results

1. Page Loading and Navigation

- **Test Case:** Verify all pages and sub-pages load correctly without 404 or missing resource errors.
  - **Result:** Initially, direct navigation to sub-pages (e.g., `/pricing`) resulted in 404 errors due to GitHub Pages' handling of client-side routing. This was resolved by:
    - Adding a `_redirects` file in the `public` directory with the rule `/* /index.html 200` to redirect all requests to `index.html` for client-side routing.
    - Changing `BrowserRouter` to `HashRouter` in `frontend/src/App.jsx` to ensure routing works correctly without server-side configuration.
  - **Status:** PASSED (after fixes)
- **Test Case:** Verify all navigation links redirect correctly.

- **Result:** After implementing `HashRouter`, navigation links (e.g., `Pricing`, `Login`) correctly redirect to their respective pages. The `Start Teaching Today` and `Watch Demo` buttons on the landing page also correctly navigate to the `Signup` page.
- **Status:** PASSED

2. Interactive Components

- **Test Case:** Verify interactive components (buttons, forms) function correctly.
  - **Result:** The `Signup` and `Login` forms load correctly. However, full end-to-end testing of login/signup functionality requires user interaction (entering credentials), which cannot be automated by the agent. The `Dashboard` page correctly redirects unauthenticated users to the login page, indicating proper protected route handling.
  - **Status:** PARTIALLY PASSED (manual verification required for full login/signup flow)
- **Test Case:** Verify audio playback and animations.
  - **Result:** No explicit audio playback elements were found on the tested pages (landing, login, signup). Animation effects (e.g., route transitions) appeared smooth during navigation.
  - **Status:** N/A (for audio, as no elements found); PASSED (for animations)

3. Responsive Design

- **Test Case:** Verify responsive design across different device sizes.
  - **Result:** The agent attempted to simulate a mobile viewport, but direct control over browser window resizing for responsive testing is not fully supported by the current tools. Visual inspection at the default viewport size suggests a responsive layout, but comprehensive testing across various device sizes would require manual verification.
  - **Status:** PARTIALLY PASSED (manual verification recommended)

4. Backend Integration (Firebase)

- **Test Case:** Verify Firebase configuration.
  - **Result:** The provided Firebase configuration was successfully integrated into `frontend/src/lib/firebase.js`. This update was deployed. Actual backend functionality (user authentication, data storage) requires manual login/signup and interaction with protected routes.
  - **Status:** PASSED (configuration updated and deployed)

Issues Found and Remediation

1. **Issue:** 404 errors on sub-page navigation (e.g., `/pricing`).
   - **Cause:** GitHub Pages' default behavior for static sites does not support client-side routing without specific configurations.
   - **Remediation:**
     - Created `frontend/public/_redirects` file with `/* /index.html 200`.
     - Changed `BrowserRouter` to `HashRouter` in `frontend/src/App.jsx`.
2. **Issue:** Build failure due to incorrect `lucide-react` icon import (`Spellcheck` vs `SpellCheck`).
   - **Cause:** Typo in the import statement for a `lucide-react` icon in `GroupView.jsx`.
   - **Remediation:** Corrected `import { Spellcheck } from 'lucide-react';` to `import { SpellCheck } from 'lucide-react';` in `frontend/src/pages/GroupView.jsx`.
3. **Issue:** Firebase configuration was outdated.
   - **Cause:** The `firebase.js` file contained placeholder or incorrect Firebase project details.
   - **Remediation:** Updated `frontend/src/lib/firebase.js` with the provided Firebase configuration details.

Conclusion

The Ladder Lessons website has been successfully deployed to GitHub Pages at `https://sharichung.github.io/ladder-lessons/`. Key issues related to client-side routing and build errors have been identified and resolved. The website's core navigation and public-facing pages are functioning as expected. Full verification of user authentication, subscription systems, and game functionalities requires manual testing due to the nature of these features and the limitations of automated browser tools for sensitive user interactions.

Next Steps

- **Manual UAT:** It is recommended that a human user perform manual UAT for:
  - Full user registration and login flow.
  - Subscription process (if applicable).
  - Interactive game functionalities (Matching Game, Spelling Practice Game).
  - Comprehensive responsive design testing across various devices.
- **Monitoring:** Continue monitoring GitHub Actions for future deployments.